

Technology Trend Analytics Platform

A step-by-step guide to building, documenting, and publishing the project on GitHub

Overview

This project analyzes search interest trends for technologies like Artificial Intelligence, Blockchain, Quantum Computing, 5G, and Cybersecurity using Google Trends data. It combines data collection, visualization, regional analysis, and machine learning forecasting (Prophet) into one platform, built entirely in Google Colab.

Features

- 5-year trend comparison across multiple technologies
- Country-wise regional interest analysis
- Word cloud of rising and related search queries
- 6-month future forecasting using Facebook Prophet
- Growth scorecard ranking technologies by momentum
- Automatic CSV export of all results

Tech Stack

Tool / Library	Purpose
Python	Core language
Pytrends	Google Trends API wrapper
Pandas / NumPy	Data processing
Matplotlib / Seaborn	Static visualizations
Plotly	Interactive charts
Prophet	Time-series forecasting
WordCloud	Rising query visualization
Google Colab	Development environment

Step-by-Step Setup Guide

Step 1: Open Google Colab

Go to colab.research.google.com, click New Notebook, and rename it to Technology_Trend_Analytics.ipynb.

Step 2: Install Required Libraries

Run this in the first cell:

```
!pip install pytrends pandas matplotlib seaborn plotly prophet requests wordcloud --quiet
```

Step 3: Import Libraries

Import all required packages for data collection, processing, and visualization:

```
from pytrends.request import TrendReq
import pandas as pd, numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px, plotly.graph_objects as go
from prophet import Prophet
from wordcloud import WordCloud
```

Step 4: Collect Google Trends Data

Connect to Google Trends and pull 5 years of interest data for chosen keywords:

```
pytrends = TrendReq(hl='en-US', tz=330)
tech_keywords = ['Artificial Intelligence', 'Blockchain',
                 'Quantum Computing', '5G', 'Cybersecurity']
pytrends.build_payload(tech_keywords, cat=0, timeframe='today 5-y')
trend_data = pytrends.interest_over_time()
```

Step 5: Visualize Trends

Use Plotly to plot interest over time for each keyword as an interactive line chart.

Step 6: Regional Analysis

Identify which countries search each technology the most:

```
region_data = pytrends.interest_by_region(resolution='COUNTRY', inc_low_vol=True)
```

Step 7: Rising Queries and Word Cloud

Pull related/rising queries and generate a word cloud to spot emerging sub-trends:

```
related = pytrends.related_queries()
```

Step 8: Forecast Future Trends

Train a Prophet model on the historical data and predict the next 6 months of interest.

Step 9: Build the Scorecard

Rank each technology by average interest and growth percentage to see what is rising fastest.

Step 10: Export Results

Save all outputs as CSV files for reporting:

```
trend_data.to_csv('tech_trends_over_time.csv')
region_data.to_csv('tech_trends_by_region.csv')
```

Step 11: Upload to GitHub

Create a new repository (e.g. Technology-Trend-Analytics-Platform), download the notebook from Colab (File > Download > Download .ipynb), then upload the notebook, this document, and the exported CSV files to the repository and commit.

Project Structure

```
Technology-Trend-Analytics-Platform/
|-- Technology_Trend_Analytics.ipynb    # Main Colab notebook
|-- README.md                          # Documentation
|-- tech_trends_over_time.csv          # Exported trend data
|-- tech_trends_by_region.csv          # Exported regional data
|-- tech_trend_scorecard.csv           # Exported growth scorecard
```

Future Improvements

- Add more data sources (GitHub Stars trends, Stack Overflow tags)
- Deploy as a live web dashboard using Streamlit or Flask
- Automate weekly trend updates via scheduled runs

Built as a portfolio project demonstrating skills in data collection, visualization, and predictive analytics.